

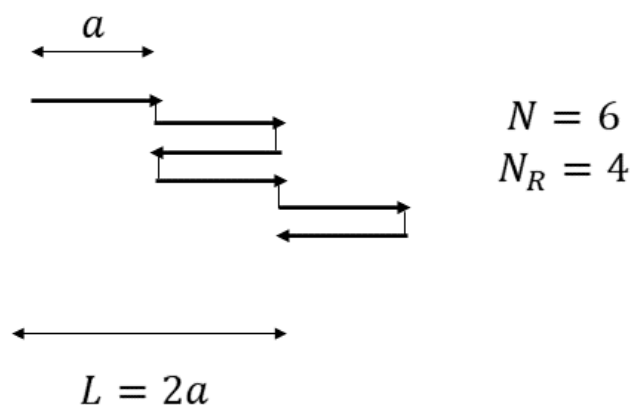
PC2135 Thermodynamics and Statistical Mechanics

Midterm test, 2 Mar 2023, 10:00-11:40 AM (10 marks for each sub-questions)

1. Consider ideal gas expanding quasi-statically and adiabatically.
 - a. What is an adiabatic process? Give the condition of an adiabatic process based on the first law of thermodynamics.
 - b. Show that when an ideal gas expands adiabatically, the temperature T and pressure P are related by the differential equation $\frac{dT}{dT} = \frac{2}{f+2} \frac{T}{P}$, here f is the degrees of freedom of one atom of the ideal gas.

2. Consider an Einstein solid with $N = 3$ oscillators and $q = 4$ units of energy packets. The energy unit is ϵ .
 - a. Draw all the pictures representing the microstates with the constraints of three oscillators and four units of energy. How many of them (i.e. the value of multiplicity Ω)?
 - b. What is the entropy S of the system? Give an approximate estimate the value of the temperature T in units of ϵ/k .

3. Polymer, such as rubber, are made of very long molecules, usually tangled up in a configuration that has a lot of entropy. As a very crude model of a rubber band, consider a chain of N links each of length a . See figure below. Imagine that each link has only two possible states represented by the arrows in the figure, pointing either left or right. The total length L of the rubber band is the net displacement from the beginning of the first link to the end of the last link.
 - a. Find an expression for the entropy of this system in terms of the total number of links N , and N_R , the number of links pointing to the right.
 - b. Write down a formula for the total length L in terms of N and N_R .
 - c. For a one-dimensional system as this, the length L is analogous to the volume V of a three-dimensional system, and the tension force is like the pressure P . Using the thermodynamics identity (suitably generalized for the current problem), determine the tension force F in terms of the variables L , T , N , and a .



4. The entropy of an Einstein solid in the thermodynamic limit is given by $S(U, \epsilon) = kN \left(\left(\frac{U}{\epsilon N} + 1 \right) \ln \left(\frac{U}{\epsilon N} + 1 \right) - \frac{U}{\epsilon N} \ln \frac{U}{\epsilon N} \right)$. Here, the entropy is viewed as a function of two variables, the total internal energy U , and the energy unit ϵ related to the oscillator frequency, keeping the number of particles as a constant.
- By differentiating the entropy, show that we have $dU = TdS + \frac{U}{\epsilon} d\epsilon$, where T is temperature.
 - Consider the $U - \epsilon$ diagram, as shown below. Give a proof that the adiabatic curves are given by $\frac{U}{\epsilon} = \text{const}$, and isothermal curves are given by $U = \epsilon N / \left(\exp \left(\frac{\epsilon}{kT} \right) - 1 \right)$.
 - Based on the result in part b, we can construct a Carnot engine running in a loop as shown in the diagram. Compute the heat absorbed Q_h at the high temperature T_h , and the total work W done by the engine, and then the engine efficiency $e = W/Q_h$.

